

Validation scoreboard by domain

Weak-field / local tests

Pure-scalar limit (no rotation)

Cassini (solar conjunction)

β cross-channel

LLR (lunar laser ranging)

Viking (solar conjunction)

Mercury (perihelion precession)

Strong-field / compact objects

GW polarization modes

Strong field (joint higher-order)

EHT (shadow)

Binary pulsar (orbital decay)

GW (GW150914 etc.)

GW (GW250114)

XRISM

Cosmology / large scale

Background metric (case A)

Cosmology (Pantheon+ Hubble)

Cosmology (distance duality)

Cosmology (cluster-collision offset)

Galaxy rotation curves (SPARC)

BBN (early thermal history)

Cosmology (CMB polarization phase)

Background metric (case B)

Cosmology ($f\sigma_8$)

Cosmology (CMB acoustic peaks)

Reference / auxiliary

GPS clocks

Velocity saturation δ (theory)

Light deflection (Sun)

Rotation (frame dragging)

GP-A / Galileo

Normalized score (0 ideal, 1 pass boundary, 2 watch boundary)